Event-TimeRAF: Event-Aware Retrieval-Augmented Foundation Model for Explainable Air Quality Forecasting Under Concept Drift

Author Name

***Abstract*—Meteorological variability, calendar effects, environmental events and distribution shift complicate short-horizon PM2.5 forecasting. While time-series foundation models offer transferable forecasting capabilities, numerically similar time-series windows may not share the local conditions that control urban air quality. TimeRAF addresses a related problem by augmenting a frozen forecaster with retrieved time-series context. This study introduces Event-TimeRAF, an event-aware retrieval framework consisting of a Source-Audited Context Builder, a Leakage-Safe Event-Context Retriever and a Drift-Aware Forecast and Evidence Head. The framework aligns official air-quality, weather, calendar and storm-event records; restricts retrieval to training-history windows whose targets precede the query lookback; evaluates feature-level XGBoost and forecast-level frozen Chronos-Bolt fusion; and stores retrievable explanation evidence. Evaluation uses EPA AirData PM2.5, NOAA Global Hourly weather and hash-checked NOAA Storm Events data from 2019 to 2024 for Los Angeles County, with a 168-hour lookback and a 24-hour horizon. The best pre-specified supervised context baseline achieves MSE 26.185, MAE 3.125, RMSE 5.117 and R^2 0.379, while the full feature-level Event-TimeRAF model achieves MSE 26.712. Frozen Chronos-Bolt achieves MSE 28.941, and its retrieval-fused variant achieves MSE 28.709, although the paired confidence interval spans zero. These values are recalculated from manifest-backed prediction artifacts for 7,199 test origins. The results indicate selective rather than uniform retrieval gains. Because Storm Events does not provide publication timestamps, the event-conditioned findings are retrospective sensitivity results rather than real-time forecasts.**

***Index Terms*—Air quality forecasting, PM2.5, retrieval-augmented forecasting, time-series foundation models, TimeRAF, concept drift and explainable forecasting.**

I. Introduction

Fine particulate matter (PM2.5) is one of the key urban air quality risks, due to its relation to health effects on respiration and cardiovascular system [1]. Correct short term PM2.5 prediction can be used for the purpose of public health warning, exposure avoidance, transportation scheduling and environmental management. However, the dynamics of PM2.5 is challenging to model based on pollutant history in Los Angeles County. Concentrations may vary depending on wind speed, boundary layer behavior, rain events, season, holidays, wildfire smoke and/or on unusual storm or high wind events. These factors means that the forecasting problem is time and context dependent.

Recent time-series foundation models (TSFMs) try to enhance the generalization ability of forecasting by learning from a huge amount of time-series data and transferring the forecasting capability to a new domain. Some models are available, such as TimesFM, Moirai, MOMENT, Chronos and Tiny Time Mixers [2]–[6]. Concurrently, it has been demonstrated that non-parametric memory can be useful for foundation models to utilize a relevant external knowledge at inference time [7], [8] through retrieval-augmented generation. In order to merge these two ideas for zero-shot TSF, TimeRAF retrieves relevant historical sequences from a curated knowledge base and use Channel Prompting [9] to inject them to a frozen TSFM.

The domain-specific retrieval question is not fully answered by numerical time-series retrieval alone. A historical PM2.5 window can be a similar window to the current one, but can be related to different weather or event conditions. On the other hand, a window having moderate numerical similarity might be more informative if having high wind, rain, holiday, smoke or stagnation context. This encourages forecasting to be done within a retrieval system and to consider source records, weather, calendar data and distribution-shift indicators as components of the forecasting evidence instead of as metadata.

In this study, we propose a new event-aware retrieval augmented framework (Event-TimeRAF) for explainable PM2.5 forecasting under distribution shift. The method takes the concept of retrieval from TimeRAF, and adapts it to an official-source environmental setting, but does not attempt to duplicate the learned retriever or Channel Prompting mechanism of TimeRAF. Event-TimeRAF is divided into three named parts: a Source-Audited Context Builder (SACB), a Leakage-Safe Event-Context Retriever (LSER), and a Drift-Aware Forecast and Evidence Head (DFEH). These components aim to answer three questions: how to set up official environmental records for retrieval; how to choose historical candidates without temporal leakage; and when retrieval can help in addition to a good supervised weather-calendar baseline.

The main contributions are listed below:

- A reproducible mechanism, called the Source-Audited Context Builder, that maintains consistency between EPA PM2.5, NOAA weather, calendar, and hash-verified NOAA Storm Events records, and maintains source identity, coverage checks, and the assumption of event-availability.
- *Leakage-Safe Event-Context Retriever:* a mechanism that combines the similarity of PM2.5 shape, weather, calendar and event similarity while enforcing an embargo, where under each candidate target ends before the look-back of the query begins.

- *Drift-Aware Forecast and Evidence Head:* Evaluation layer that provides direct XGBoost forecast and retrieval fusion based on the same chronological protocol as Chronos-Bolt, transparent shift indicators and machine-readable explanation evidence.

The scoring is done very conservatively. It only compares models when they are manifest-backed; it doesn't bias interpretation of event association to causation, and it only shows uncertainty for paired comparisons. The results hence provide a description of the situations in which retrieval is beneficial, and those in which more evidence is needed before it can be said that operational event-aware forecasting is possible.

II. Literature Review and Related Work

A. Statistical, Machine-Learning, and Deep Forecasters

Forecasting techniques differ in their inductive biases, data requirements and computational costs. Autoregressive and seasonal models are informative baselines because their failure modes are readily inspected. Tree ensembles (e.g., XGBoost and LightGBM) build on this configuration by supporting nonlinear interactions among lag, rolling, weather and calendar features without requiring a sequence encoder [10], [11]. Recurrent LSTM and GRU models, however, learn stateful temporal representations [12], [13]. Reproducible comparison across heterogeneous time-series domains is also supported by large benchmark archives such as the Monash Forecasting Repository [14]. Along this line, transformer families model longer dependencies: Informer reduces attention cost; Autoformer and FEDformer decompose time series; Crossformer and iTransformer model cross-variable structure; and TimesNet rearranges temporal variation into two-dimensional patterns [15]–[20]. These benefits do not necessarily mean that a more complex architecture is always required. Simple decomposition- or patch-based inductive biases remain competitive in DLinear and PatchTST [21], [22], while TimeMixer explicitly decomposes and mixes information at multiple scales [23]. Thus, Event-TimeRAF does not assume a deep backbone to be the default winner, but retains seasonal and boosted-tree baselines.

B. Time-Series Foundation Models

Time-series foundation models (TSFMs) aim to transfer across datasets as opposed to training a new representation from scratch for every time series. The former is a decoder-only forecasting design (TimesFM), the latter a universal forecasting Transformer (Moirai), the next one supports multiple time-series tasks (MOMENT), the next one encodes values as tokens (Chronos), and the last one focuses on compact zero- and few-shot transfer (Tiny Time Mixers) [2]–[6]. Timer, Timer-XL, UniTime, and Time-LLM explore generative pretraining, long-context scaling, cross-domain unification, or language-model reprogramming [24]–[27]. It was also found that frozen pretrained models can be used as general computation engines for analyzing time series [28]. All these studies encourage zero-shot evaluation, and also indicate that the performance of the transfer also relies on input representation and domain shift. Therefore, in the present study, we do not want to assume that the models based on a foundation model are better, and we want to compare frozen Chronos-Bolt and locally supervised models on the same forecast origins.

C. Retrieval-Augmented Forecasting

Retrieval augmentation separates parametric knowledge from an external memory. The retrieve-then-condition pattern was pioneered by RAG and Dense Passage Retrieval in NLP [7], [8]. In the context of time series, retrieval has been investigated in diffusion forecasting and assisted-generation workflows [29], [30]. TimeRAF is the most similar method: it builds a time-series knowledge base, uses an MLP dual-encoder retriever, finds the top k candidates and incorporates candidate embeddings into a frozen TSFM through Channel Prompting [9]. Its ablations examine retrieval quality, integration, the number of candidates, knowledge-base composition and backbone selection.

Event-TimeRAF is based on the same external-memory principle but is designed to answer a different experimental question. It uses transparent random, cosine and event-context similarity instead of a learned dual encoder, and feature- or forecast-level fusion instead of internal Channel Prompting. The current implementation is therefore an audited environmental adaptation intended to determine whether contextual historical analogues assist a local PM2.5 forecasting task, rather than a reproduction of TimeRAF.

D. Air-Quality Forecasting and Official Context

History of pollutants along with temperature, humidity, wind, precipitation and pressure are commonly used in PM2.5 forecasts. In recent years, there have been reviews of single-station recurrent models to attention-based, hybrid, and spatiotemporal models [31], [32]. For example, AirFormer models at a nationwide level by separating the deterministic spatiotemporal modeling from the stochastic uncertainty estimation [33]. These multi-station designs account for spatial dependence, while the current study considers a county-level aggregate, and a source of contextual evidence. Observed PM2.5 data is provided by the EPA AirData, hourly meteorology is provided by NOAA Integrated Surface Database, and structured hazard data is provided by NOAA Storm Events [34]–[36]. Evidence of smoke could be included in NOAA Hazard Mapping System products, but were not used in the run reported [37].

E. Distribution Shift and Forecast Explanations

Temporal nonstationarity may lead to changes in means, variances and relationships between inputs and future values. Adaptive normalization methods address this problem by estimating statistics at local temporal scales [38]; Event-TimeRAF instead uses interpretable indicators of shift to audit regimes without claiming that shift has been eliminated. There are also varying degrees of scope of explanation. SHAP can be used to attribute predictions to model inputs [39] and information-theoretic saliency can be used for counterfactual inspection of probabilistic forecasts [40]. The explanation layer implemented is more restricted: it only captures XGBoost signed effects, retrieved cases, event identifiers, drift components, and an uncertainty proxy. These records help with the traceability, however, they do not help to infer causal relationships between weather and events.

III. Critical Gaps and Limitations of Previous Studies

Existing studies leave three gaps that need to be addressed in the context of event sensitive urban air-quality forecasting. First, temporal forecasters and temporal numerical retrieval systems do not necessarily take care of the provenance, coverage, and availability assumptions of local contextual records. A numerically similar PM2.5 window may happen during a weather regime or event different from that indicated by the retrospective event label, and a retrospective event label could be misinterpreted as information at forecast time. The SACB aims to fill this void by incorporating official records through explicit source, coverage and availability audits.

Second, historical retrieval may include look-ahead bias if there is overlap between the candidate input and the query context, or between the candidate target and the query context. The fact that it is similar doesn't mean that a near duplicate future doesn't get into the retrieved evidence. The LSER resolves this with its training-history restriction and strict temporal embargo, followed by PM2.5, weather, calendar and event similarity terms, which can be audited independently.

Third, retrieval gain is sometimes reported without evaluation of whether there is already a powerful local context model that captures the useful signal, whether retrieval gain still exists under distribution shift, and whether there is evidence to explain retrieval gain. The DFEH fills this gap by conducting assessments of both the direct XGBoost and frozen Chronos-Bolt paths on common origins, highlighting shift indicators, and maintaining machine-readable feature, retrieval, event and uncertainty evidence. This is a component of comparison and auditability; causal explanations are not produced.

These mappings form the boundaries of the novelty of the study. Event-TimeRAF does not have a learned dual-encoder retriever or internal Channel Prompting, and the current experiment is only done on a single county, without establishing geographic generalization.

IV. Core Contributions

The proposed framework provides three mechanisms, that correspond one on one with the three gaps listed above:

- *Source-Audited Context Builder (SACB):* builds an aligned context that is 85 dimensions based on official PM2.5 and meteorological data, calendar and event information, and retains source and availability metadata.
- *Leakage-Safe Event-Context Retriever (LSER):* retrieves training-history analogues only after implementing a candidate-target embargo and provides exposure of the contribution of each similarity channel and retrieved trajectory evidence.
- *Drift-Aware Forecast and Evidence Head (DFEH):* integrates all of the above into a single chronological evaluation contract.

TABLE I

Main Symbols Used in the Proposed Formulation

Symbol	Meaning
L	Lookback window length
H	Forecast horizon
X_t	Historical PM2.5 sequence at time t
W_t	Weather covariates
C_t	Calendar/context features
E_t	Event context available at the forecast origin
R_t^{ts}	Retrieved time-series knowledge
R_t^{ev}	Retrieved event knowledge
D_t	Drift indicators
q_t	SACB context vector in $\mathbb{R}^{85}$
G_t	LSER top-k retrieved set
$\rho(G_t)$	Retrieval summary in $\mathbb{R}^{51}$
$\mathcal{O}_t$	Machine-readable DFEH evidence record
$\hat{Y}_{t+1:t+H}$	Predicted future PM2.5 values

The empirical contribution is intentionally qualified: weather-calendar context produces the clearest supported gain, while event and retrieval components do not produce uniformly positive improvements.

V. Problem Formulation

Let x_t be the measurement of PM2.5 at time t. The lookback window for the air quality is defined by Equation (1) for forecast time t.

$$X_t = [x_{t-L+1}, x_{t-L+2}, \dots, x_t], \tag{1}$$

Here, L is the lookback length, W_t contains weather covariates aligned with the lookback window, C_t contains calendar features, and E_t contains event-context variables available at the forecast origin under the recorded source assumption. The target in Equation (2) corresponds to the input in Equation (1).

$$Y_{t+1:t+H} = [x_{t+1}, x_{t+2}, \dots, x_{t+H}], \tag{2}$$

where H represents the horizon of forecast. Equation (2) thus represents the 24-step vector for each allowable origin.

The forecasting function is given by Equation (3):

$$\hat{Y}_{t+1:t+H} = F_\theta(X_t, W_t, C_t, E_t, R_t^{ts}, R_t^{ev}, D_t), \tag{3}$$

where R_t^{ts} is a retrieved time-series knowledge, R_t^{ev} is an event knowledge retrieved and D_t is a concept-drift indicator set. The target setting which has been implemented is: L = 168 hours, H = 24 hours. The full information interface is given in Equation (3) and the following section describes the modules that are implemented to build the arguments.

The time based split is given by Equation (4).

$$\mathcal{T}_{train} < \mathcal{T}_{val} < \mathcal{T}_{test}, \tag{4}$$

Here the first 70% of observations are employed to train the system, the second 15% for validation and the final 15% for testing. The order in (4) is preserved for feature fitting, retrieval eligibility, model selection and final evaluation.

The notation used in the rest of the report is summarized in Table I.

VI. METHODOLOGY

The Event-TimeRAF is composed of three main components. The Source-Audited Context Builder (SACB) provides a way to translate from the heterogeneous official records to a chronological hourly context while preserving the provenance and availability metadata. The Leakage-Safe Event-Context Retriever (LSER) builds the embargoed historical memory as well as retrieves the analogues based on PM2.5 similarity, weather similarity, calendar similarity and event similarity. The Drift-Aware Forecast and Evidence Head (DFEH) reveals feature level and forecast level prediction paths, prediction shift indicators, and a machine readable explanation record. No image operator, learned attention block, or internal TimeRAF Channel Prompting is implied with these components which are deterministic or supervised modules over one dimensional time-series and tabular inputs.

A. Framework Overview and Forward Pass

The whole dataflow is summarized in figure 1. The first data to enter the SACB are EPA PM2.5, NOAA weather, calendar and NOAA Storm Events records that are audited and aligned then transformed into a query context and input-target windows. The LSER filters the knowledge base of training-histories, ranks candidates eligible for the filter and returns the retrieved candidate futures and retrieval statistics. The DFEH then writes those statistics to 24 direct regressors of XGBoost, evaluates the separate frozen fusion path of Chronos-Bolt, calculates drift evidence and writes the forecast and supporting record of the forecast.

The aligned Event-TimeRAF context is mapped to a forecast, via Equation (5), for each forecast origin t.

$$\hat{Y}_{t+1:t+H} = F_\theta\big(\phi(X_t, W_t, C_t, E_t), G_t, D_t\big), \tag{5}$$

In which, phi(.) is the tabular feature construction, G_t is the retrieved historical evidence and D_t is the drift evidence. A major limitation of the method is related to the availability of all the components of G_t and D_t at or before the time of the forecast. The tensor level forward pass below is the expanded form of Equation (5).

In tensor notation, Equation (6) expands the forward pass that was implemented.

$$\begin{aligned}
Q &= \mathrm{SACB}(X, W, C, E) && \in \mathbb{R}^{B\times 85},\\
G &= \mathrm{LSER}(Q, \mathcal{K}) && \in \mathbb{R}^{B\times 8\times 24},\\
U_h &= [Q \,\|\, \rho(G) \,\|\, d \,\|\, C_h^+] && \in \mathbb{R}^{B\times 151},\\
\hat{Y}^X &= [f_1(U_1), \ldots, f_{24}(U_{24})] && \in \mathbb{R}^{B\times 24},\\
(\hat{Y}, \mathcal{O}) &= \mathrm{DFEH}(\hat{Y}^X, \hat{Y}^C, \hat{Y}^R, d, G) && \in \mathbb{R}^{B\times 24} \times \mathcal{E},
\end{aligned} \tag{6}$$

where Q is the origin context (85-dimensional), $\rho(G) \in \mathbb{R}^{B\times 51}$ contains the retrieved trajectory, spread, two similarity summaries and the number of candidates, $d \in \mathbb{R}^{B\times 6}$ are five drift components and their mean score, and $C_h^+ \in \mathbb{R}^{B\times 9}$ is the known calendar vector for the horizon h. The symbol E denotes the schema of the evidence record O. Equation (6) describes both reported forecasting paths without presenting the XGBoost path as part of the frozen foundation model.

B. Source-Audited Context Builder

The motivation behind the SACB is to maintain the data provenance and temporal availability prior to model fitting. Figure 2 illustrates that source specific checks are done first and then the feature is built. EPA monitor records are aggregated to an hourly county median, NOAA weather data from the selected surface station are matched up to the same time, event information is hash verified against an event manifest and calendar information is calculated from the local time. A windowed and chronologically split context is thus generated, no target value is ever interpolated.

1) Dataset Construction: The hourly observations of PM2.5 are obtained from the US EPA/AirData/AQS parameter 88101 (Los Angeles County) [34]. The data audit chooses one monitoring site based on coverage and continuity criteria, if there is one monitoring site that meets the coverage criterion. Since no particular monitor met this criterion, the target is the county median measured by the official Los Angeles County AQS monitors for every hour. Times are to the `America/Los_Angeles` time zone, daylight saving time is handled explicitly. Targets are not interpolated, short input gaps can only be filled with past observations. Weather covariates are obtained from the closest weather station from the NOAA Integrated Surface Database (ISD) [35] that is suitable. Calendar elements include cyclic hour, weekday and month terms, weekend, US federal holiday and California holidays.

The feature groups are merged at all the origins using Equation (7).

$$q_t = [p_t \,\|\, w_t \,\|\, c_t \,\|\, e_t] \in \mathbb{R}^{23+14+9+39} = \mathbb{R}^{85}, \tag{7}$$

Here the feature vectors p_t, w_t, c_t, and e_t are the PM2.5 feature vector, weather feature vector, calendar feature vector and event feature vector. The PM2.5 group consists of fixed lags, rolling moments and extrema, differences, the current value and a fill flag. The weather group contains six measured variables, a precipitation-missing flag, derived condition flags, a stagnation proxy, and 24-hour running means. Event group consists of group of 1 hour counts, group of active counts, 24/72/168 hour counts, category specific 72 hour counts and event-burst ratio. Therefore, Equation (7) provides the ordering of features as well as the 85 dimensional SACB output to be fed to downstream modules.

The statistics for normalization are only fit on the training split. Both query and candidate windows are normalized in the same way, based on the "TimeRAF" principle that sequences retrieved and input must follow the same way of normalization. The reported run contains 48,332 valid windows with $X \in \mathbb{R}^{48332\times 168}$ and $Y \in \mathbb{R}^{48332\times 24}$.

C. Leakage-Safe Event-Context Retriever

The LSER differentiates between temporal eligibility and relevance. First, the embargo is shown (Figure 3): if the entire target of the embargo is before the start of the query lookback then the candidate is considered. The selected candidates are then compared on four similarity channels which are independently stored. Selected futures are on the same scale

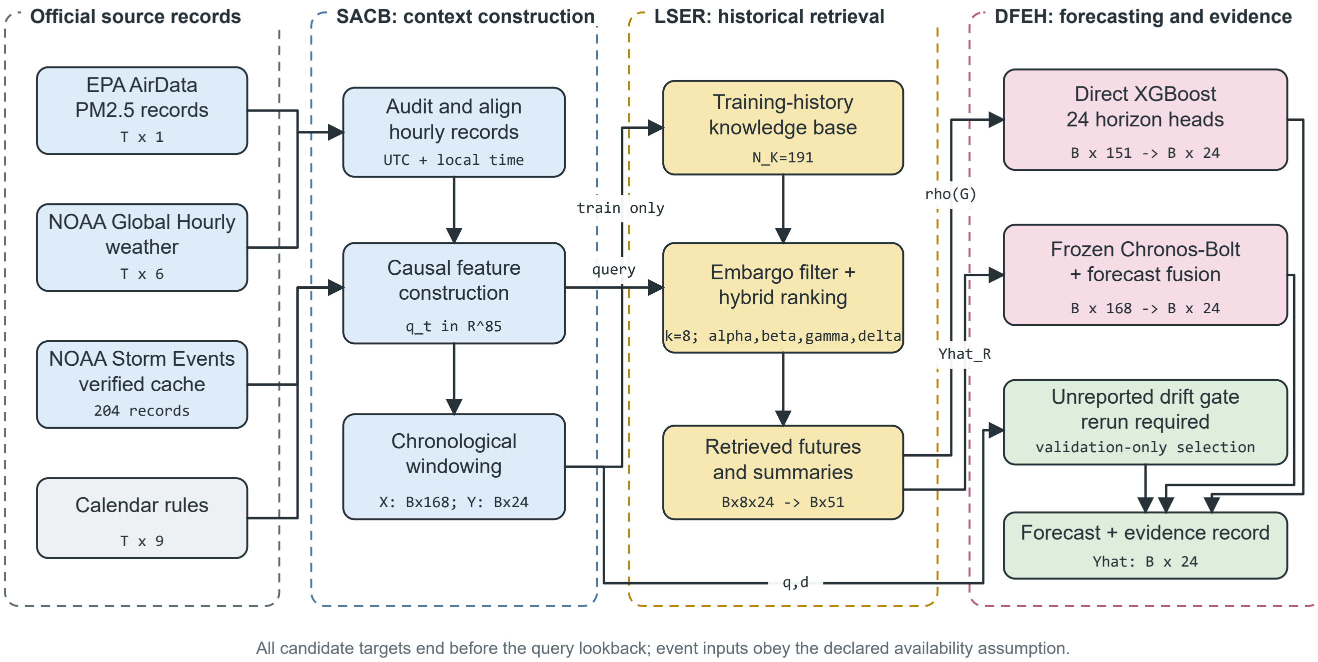

Fig. 1. Proposed Event-TimeRAF pipeline. Official records pass through the Source-Audited Context Builder (SACB), training-history candidates are selected by the Leakage-Safe Event-Context Retriever (LSER), and the Drift-Aware Forecast and Evidence Head (DFEH) produces 24-hour forecasts and traceable evidence. Tensor annotations show the reported configuration.

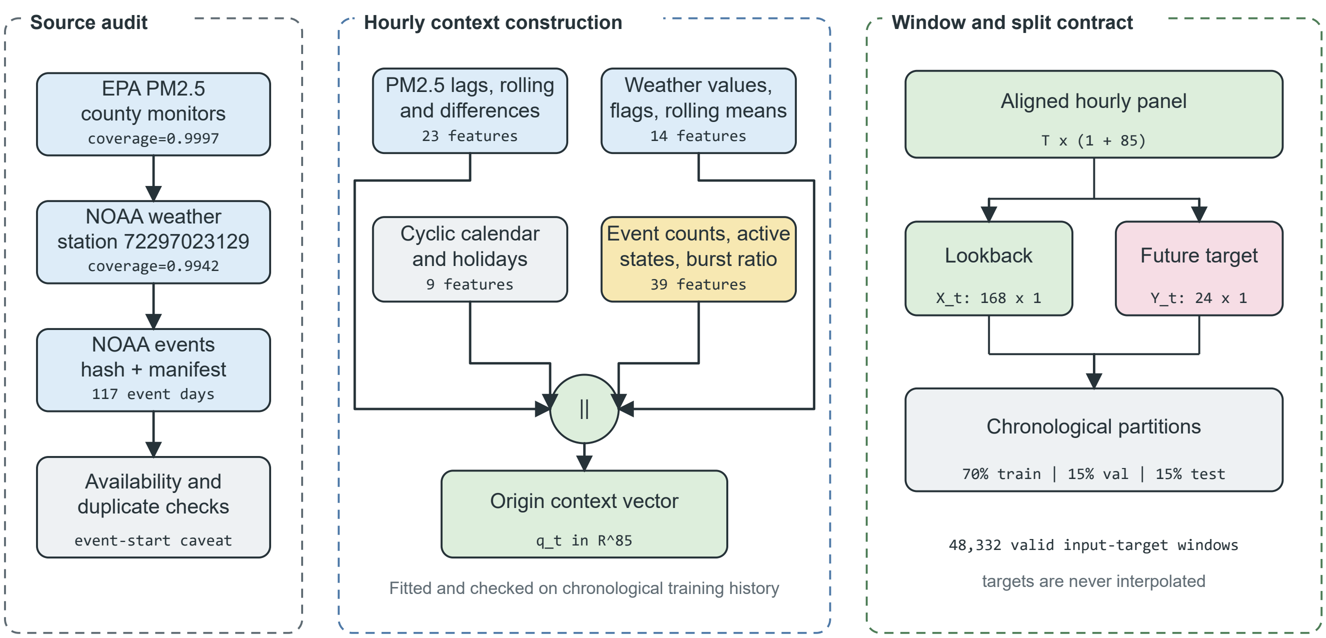

Fig. 2. Source-Audited Context Builder. Source checks feed four feature groups that form an 85-dimensional origin vector. The aligned panel is converted into 168-hour inputs and 24-hour targets before the chronological split.

as the query, and summarized for the forecasting head. This ordering is to ensure that a high similarity score does not overrule the temporal constraint.

1) Time-Series Knowledge Base: The historical PM2.5 windows are included in the time-series knowledge base. Records will save a window identifier, start time, end time,

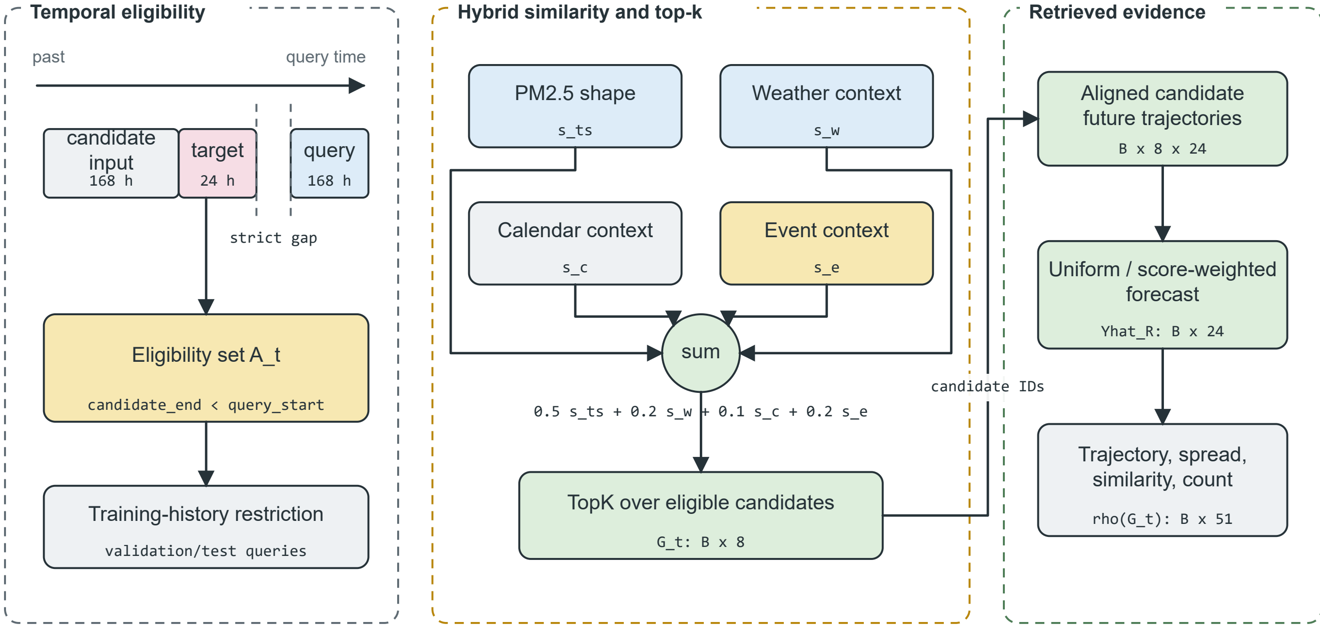

Fig. 3. Leakage-Safe Event-Context Retriever. A strict candidate-target embargo and training-history restriction are applied before hybrid ranking. The top eight candidates provide aligned future trajectories and a 51-dimensional retrieval summary.

input and future values, weather summary, calendar summary, normalized window vector, split and a normalization identifier. The 192 hours of separation between knowledge-base origins, is composed of the 168 hours of input and 24 hours of future. This reported knowledge base has 191 training windows which are not overlapping.

Equations (8) formally defines the knowledge base.

$$\mathcal{K} = \{(u_i, X_i, Y_i, m_i, c_i, e_i)\}_{i=1}^{N}, \tag{8}$$

In this case, u_i are the list of candidate origins, X_i and Y_i are the candidate lookback and future windows, m_i the list of weather summaries, c_i the list of calendar summaries and e_i the list of event-context summaries available at u_i. The set of temporally eligible origins is defined as in Equation (9) for a query origin t.

$$\mathcal{A}_t = \{i : u_i + H < t - L + 1,\ i \in \mathcal{T}_{train}\}, \tag{9}$$

To validate and test queries. The strict inequality in (9) means that a candidate's target period ends before the query period is input. The training-history condition for (9) also applies to the final validation and test retrieval. These two schemas, stored and admissible at query time, are defined in Equation (8) and Equation (9), respectively.

The time-series retriever returns the records defined by Equation (10), for a given query window X_t.

$$R_t^{ts} = \{r_{t,1}^{ts}, r_{t,2}^{ts}, \ldots, r_{t,k}^{ts}\}, \tag{10}$$

Here k is the number of retrieved candidates. The default is k=8 and sensitivity analysis of k in $\{1, 4, 8, 16\}$ is performed. Equation (10) is for the records returned before the future trajectory of the records is summarized.

2) Event Knowledge Base: The event knowledge base provides source preserving records, which relate to Los Angeles County. NOAA Hazard Mapping System fire and smoke records are still considered an unmeasured optional extension [36], [37] while NOAA Storm Events are the required structured source. Event identifier is stored, as well as start, end, information-availability assumption, location, category, and source and source URL, title, and summary. Reported records includes wildfire, high wind, heavy rain, flood and other categories of weather; empty categories for missing weather types are set to zero.

Model inputs consist of hourly counts, 24, 72, and 168 hr rolling counts, 72 hr counts broken down into event categories, and active-event counts and an event-burst ratio. Explanation evidence is kept for the last recent source identifiers. The target-horizon overlap is only considered as a post hoc subset label and is not provided as a model input. The publication time is not directly provided by NOAA Storm Events; event start is recorded as the retrospective availability assumption. The records must have real publication or issue dates for strict operational experiments.

3) Hybrid Ranking and Future Alignment: The query and candidate PM2.5 windows are normalized in both their means and standard deviations, and unit length. The candidate statistics for the weather and calendar groups are standardized and mapped to [0,1] by using the cosine similarity. Handle event similarity with clipping to [0,1] and handling of a zero vector. The hybrid retrieval score, which incorporates time-series similarity, weather relevance, calendar relevance and event relevance is given by Equation (11).

$$s(q,i) = \alpha s_{ts}(q,i) + \beta s_w(q,i) + \gamma s_c(q,i) + \delta s_e(q,i), \quad (11)$$

where, s_{ts} is time-series similarity, s_w is weather similarity, s_c is calendar similarity and s_e is event relevance. The weights that were used are: $\alpha = 0.5$, $\beta = 0.2$, $\gamma = 0.1$ and $\delta = 0.2$. The relevance of each channel of the equation is kept explicit to make it possible to analyze the contribution of each relevance channel without the need for a learned latent scoring function as in (11).

The set to be retrieved is chosen based on Equation (12).

$$G_t = \mathrm{TopK}_{i \in \mathcal{A}_t} \; s(t,i). \quad (12)$$

The only evaluation is done for equation (12) once set $\mathcal{I}_t$ is applied (which is only when $\mathcal{I}_t$ is eligible). The candidate futures are reinstated in the query scale in equation (13).

$$\widetilde{Y}_{t,i} = \mu_t + \sigma_t \left(\frac{Y_i - \mu_i}{\max(\sigma_i, \epsilon)} \right), \qquad i \in G_t, \quad (13)$$

where (μ_t, σ_t) and (μ_i, σ_i) are the statistics of the query and the candidates to look back respectively. Forecasts for retrieval only are made by taking an average of the aligned candidate futures, uniformly or using non-negative normalized scores. The feature-level path has the same mean trajectory, horizon-wise spread, mean and maximum time-series similarity, and number of candidates, so as to give $\rho(G_t) \in \mathbb{R}^{51}$. The affine transformation in Equation (13) maintains the shape of the candidate trajectory, while reverting the query trajectory's local scale.

D. Drift-Aware Forecast and Evidence Head

The common output layer of two verified forecasting paths and the evidence output is the DFEH. A direct path is defined as the concatenation of the context, retrieval, drift and future-calendar features for 24 independently-trained XGBoost regressors, as illustrated in Figure 4. The second track is a value of the retrieved forecast, mixed with the forecast of the frozen Chronos-Bolt at a validation-selected weight. The current code also includes a drill of a validation-only drift-state selector which is not represented in this manuscript in the context of a manifest-backed run.

1) Feature-Level Forecasting and Drift Gate: The direct input and prediction to the model is given by Equation (14) for horizon h.

$$u_{t,h} = [q_t \,\|\, \rho(G_t) \,\|\, d_t \,\|\, c_{t+h}], \qquad \hat{y}^X_{t+h} = f_h(u_{t,h}), \quad (14)$$

$u_{t,h} \in \mathbb{R}^{151}$ for the complete M09 model and each f_h is an XGBoost regressor trained with squared error objective. The design allows for each forecast horizon to have a supervised function, but shares a common origin representation. The verified 151 features (85 context, 51 retrieval, six drift and nine future-calendar) are also included in equation (14).

The tabular fusion stage defines a lightweight drift-conditioned selector. It identifies the strongest validation candidate separately for drift and non-drift origins before selecting the corresponding forecast source for each test origin, as indicated in Equation (15).

$$\hat{Y}^G_t = \begin{cases} \hat{Y}^a_t, & D_t = 0, \\ \hat{Y}^b_t, & D_t = 1, \end{cases} \quad (15)$$

where a and b are selected using validation errors only. The gate does not train a new model; it tests whether a transparent drift-specific rule can select a different forecaster for each regime. Because its predictions, metrics and checksums are not included in the verified manifest, this selector is excluded from the reported model comparison.

2) Frozen-TSFM Forecast Fusion: The Chronos-Bolt retrieval variant is powered with forecast level fusion, instead of inbuilt Channel Prompting. Define $\widehat{Y}^C_t$ as the "frozen" forecast of Chronos and $\widehat{Y}^R_t$ as the "retrieved" historical forecast. The fused prediction is given by equation (16).

$$\hat{Y}^{CR}_t(\omega) = \omega \hat{Y}^C_t + (1-\omega)\hat{Y}^R_t, \quad (16)$$

where omega in $\{0, 0.25, 0.5, 0.75, 1.0\}$ is selected on the validation split. Validation selected omega=0.75. This branch will not change anything on Chronos-Bolt embeddings or weights as Fusion is performed after both forecasts are made. As such, equation (16) is not a prompting operation by the TSFM, rather it is an output-space interpolation.

3) Concept-Drift Evidence: The implementation is distribution shift with transparency-indicators and not claiming that concept drift is identified perfectly. The indicators include the recent mean shift, recent variance shift, retrieval similarity drop, weather-pattern shift and event-burst ratio. The raw vector that the code works with is specified by equation (17).

$$r_t = \Bigg[|\mu_{24,t} - \mu_{168,t}|, \left| \log \frac{\sigma_{24,t} + \epsilon}{\sigma_{168,t} + \epsilon} \right|, 1 - \mathrm{clip}(\bar{s}_{ts,t}, 0, 1), \sqrt{\frac{1}{4} \sum_{\ell=1}^{4} \left(\frac{w_{t,\ell} - \widetilde{w}^{tr}_\ell}{s^{tr}_{w,\ell}} \right)^2}, \; \max(b_{event,t}, 0) \Bigg], \quad (17)$$

The weather term is based on the temperature, relative humidity, pressure and wind speed, standardized using the values of the training medians and standard deviations. The robust scaling of each component $r_{t,j}$, by the training median and training median absolute deviation (MAD), is done according to equation (18).

$$d_{t,j} = \frac{1}{6} \, \mathrm{clip} \left(\frac{r_{t,j} - \mathrm{med}_{tr}(r_j)}{\max(1.4826 \, \mathrm{MAD}_{tr}(r_j), \epsilon)}, 0, 6 \right). \quad (18)$$

Lastly, the continuous score and binary flag is defined with Equation (19).

$$S_t = \frac{1}{5} \sum_{j=1}^{5} d_{t,j}, \qquad D_t = \mathbb{I}[S_t \geq Q_{0.90}(\{S_v : v \in \mathcal{T}_{val}\})]. \quad (19)$$

Equations (17)–(19) are equivalent to the detector which was used: the reference statistics are set using training data, the threshold 0.90 is set using validation data and test data are transformed without refitting. The binary D_t is the state which is applied to Equation (15).

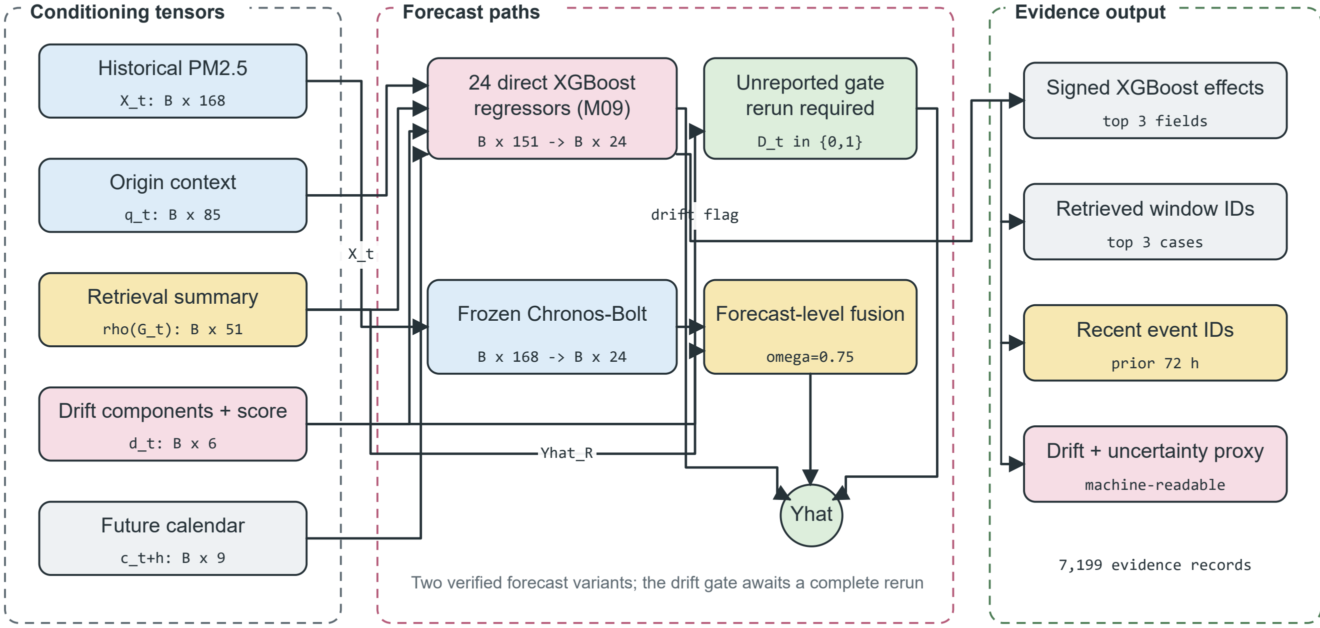

Fig. 4. Drift-Aware Forecast and Evidence Head. Context, retrieval, drift, and future-calendar tensors feed direct horizon regressors; the frozen Chronos-Bolt path is fused only at forecast level. Stored feature effects, retrieved cases, event identifiers, drift components, and uncertainty values form the evidence record.

4) Evidence Record and Uncertainty Proxy: The explanation module is based on signed local contributions provided by the 24 direct models, weather flags, top three retrieved windows, recent event records, calendar context, forecast direction, and component level drift evidence. The uncertainty proxy of it is defined by equation (20).

$$u_t^{\text{proxy}} = \sqrt{\bar{\sigma}_{R,t}^2 + \bar{e}_{val}^2}, \quad (20)$$

The proxy uses the mean absolute validation residual $\bar{e}_{\text{val}}$ and the mean horizon-wise spread of the retrieved trajectories $\bar{\sigma}_{R,t}$. It provides a diagnostic scale rather than a calibrated prediction interval. Each explanation is constructed from stored fields rather than an unrestricted language model, and every event statement is contextual rather than causal. Equation (20) therefore defines an evidence field and is not used to construct confidence intervals.

E. Relationship to TimeRAF

TimeRAF utilizes a learnable dual encoder with an MLP to retrieve top-k windows from time-series, and uses Channel Prompting [9] to inject candidate embeddings into a pre-trained TSFM. Table II shows which of the principles are kept and which of the mechanisms are changed. It is a methodological comparison not a performance ranking, since the studies are based on different sets of data, horizons and protocols.

VII. Experimental Setup

This section describes the reported experiment. For models M00 to M11, model selection was fixed using the training and validation partitions before the test results were examined. A later post-hoc selector is excluded from the reported comparisons because it was added after the verified manifest was created. Run identifier 20260723T112033170131Z was used to verify artifact consistency.

A. Dataset Description and Audit

The study period is 2019-2024. Los Angeles County PM2.5 data were collected from US EPA, AirData/AQS parameter 88101 [34]. The official Los Angeles County AQS monitors' hourly county median was used as the audited target series (not any one monitor site). This fall back picked 06-037-COUNTY and observed target hours of 52,602 and observed PM2.5 coverage of 0.9997. The PM2.5 value was always measured in micrograms per cubic meter.

Weather covariates data was downloaded from NOAA Global Hourly/ISD, using the NOAA Open Data Dissemination public buckets [35]. The selected station was 72297023129 (Long Beach / Daugherty Field / Airport) at a distance of 10.28 km to the PM2.5 target centroid. The full weather-feature coverage following past-only short-gap filling was 0.9942 and the raw measured core coverage was 0.9905.

NOAA Storm Events annual detail archives [36] were extracted and loaded as environmental events, with duplicate events removed. The experiment used a locally staged cache of unchanged NOAA annual archives and a source manifest listing the official NOAA URLs, byte count and SHA-256 hash for each archive. No delivery-mode change was required during the audit, and the `verified_official_noaa_cache` mode was retained.

TABLE II
METHODOLOGICAL COMPARISON OF TIMERAF AND EVENT-TIMERAF

Aspect	TimeRAF	Event-TimeRAF
Primary objective	Improve zero-shot forecasting by augmenting a frozen TSFM with external time-series knowledge.	Evaluate source-audited, event-context retrieval for local PM2.5 forecasting under temporal shift.
Forecasting domain	Multi-domain benchmark datasets.	Los Angeles County hourly PM2.5, 2019–2024.
Knowledge base	Curated, consistently normalized time-series windows.	191 non-overlapping training-history windows with PM2.5, weather, calendar, and retrospective event context.
Retriever	Learned MLP dual encoder with forecaster-guided training.	Random, cosine, and transparent hybrid similarity; no learned retriever.
Knowledge integration	Internal Channel Prompting of candidate embeddings.	Feature-level XGBoost inputs and forecast-level Chronos-Bolt fusion.
Backbone	Frozen TSFM, principally TTM-Base.	Direct XGBoost horizon models and frozen `amazon/chronos-bolt-small`.
Leakage control	Non-overlapping knowledge windows and cross-dataset training constraints.	Candidate target must end before query lookback; validation and test retrieval use training history only.
Event handling	Not an explicit module.	Availability-tagged NOAA Storm Events features and source-record evidence, reported retrospectively.
Drift and explanation	Retrieved-case inspection.	Five-component drift audit, signed feature effects, retrieved IDs, event IDs, and uncertainty proxy.
Implementation scope	Learned retriever and Channel Prompting are implemented.	Those mechanisms are not implemented and remain future extensions.

TABLE III
DATA-READINESS AUDIT

Gate	Value	Status
PM2.5 observed coverage	0.9997	Pass
Weather complete coverage	0.9942	Pass
Event days	117	Pass
Event overlap days	2,192	Pass
Qualifying event categories	3	Pass
Strict event availability	False	Caveat

The event audit identified 117 event days, 2,192 source-overlap days and three qualifying event categories. Because NOAA Storm Events does not provide machine-readable publication timestamps, the event-aware outputs retrospectively assume availability at the event start time and are not, by themselves, real-time event forecasts. The resulting readiness status is given in Table III. Figure 2 shows the source audit window and window contract.

Table IV summarizes the assessed scope and the various sources' contribution to the forecasting process.

B. Data and Code Availability

EPA AirData, NOAA Global Hourly/ISD and NOAA Storm Events provide the raw public records and download pages [34]–[36]. Because the local execution environment could not access NCEI directly, the experiment used a locally staged NOAA Storm Events cache, and every cached NOAA file was checked against the generated source manifest. Source code is available at https://github.com/shasabbir/event-time-raf. The implementation uses Python 3.12.13, NumPy, pandas, scikit-learn, XGBoost, PyTorch and the chronos-forecasting package. The repository contains the configuration, notebooks, source modules and tests required to reproduce the pipeline when the public source files are available.

C. Forecasting Task and Split

The forecasting task involves looking at historical information for a 168-hour period and predicting PM2.5 for the next 1-24 hours. The split is done in chronological order, the first 70% of valid windows are for training, the next 15% of valid windows for validation and the last 15% of valid windows for testing. The final test set has 7,199 forecast origins and 172,776 forecast target points. Random splitting is NOT performed.

D. Models

Forecast fusion of retrieval-augmented Chronos-Bolt and the following models are evaluated: persistence, daily seasonal naive, weekly seasonal naive, direct XGBoost model using PM2.5 features, direct XGBoost context model using PM2.5, weather, and calendar features, random retrieval, cosine retrieval, XGBoost with cosine retrieval features, Event-TimeRAF without drift features, full Event-TimeRAF with event and drift features, and frozen Chronos-Bolt. Ablation analysis is carried out with the help of an auxiliary random-retrieval model A01 and an auxiliary event-free model A00. Post-Hoc M12 artifact is not a part of the manifest-backed result set and hence is excluded. The amazon/chronos-bolt-small checkpoint is used for the foundation-model evaluation. All of the assessed configurations are categorized based on their forecasting and retrieval functions in Table V.

E. Implementation and Hyperparameters

This experiment was conducted in a cloud-based Linux 6.12.90 operating system with Python programming language (Python 3.12.13). The run time was 1,576.87 seconds. Some of the package versions listed in the run manifest are: NumPy 2.0.2, pandas 2.3.3, scikit-learn 1.6.1, XGBoost 3.2.0, PyTorch 2.10.0+cu128 and chronos-forecasting 2.3.1. The main hyperparameters are given in table VI.

TABLE IV
DATASET AND SOURCE SUMMARY

Data group	Source	Evaluated scope	Use in pipeline
PM2.5	EPA AirData/AQS hourly parameter 88101	Los Angeles County, 2019–2024; 52,602 observed target hours	Target sequence and lag/rolling features
Weather	NOAA Global Hourly/ISD, station `72297023129`	2019–2024; 0.9942 usable core coverage	Temperature, humidity, wind, pressure, and rain features
Events	NOAA Storm Events annual detail archives	204 Los Angeles records; 117 event days	Retrospective event-start features and explanation evidence
Calendar	Generated from timestamps and public holiday rules	2019–2024 hourly calendar	Hour, weekday, weekend, month, season, and holiday features

TABLE V
EVALUATED MODEL FAMILIES

ID	Description
M00–M02	Persistence and seasonal naive baselines
M03–M04	XGBoost PM2.5-only and context models
M05–M06	Random and cosine retrieval-only baselines
M07	XGBoost with cosine retrieval features
M08–M09	Event-TimeRAF without/with drift indicators
M10–M11	Frozen Chronos-Bolt and retrieval fusion

F. Evaluation Metrics and Statistical Checks

The main metrics are: MSE, MAE, RMSE, MAPE, sMAPE and R^2. Although MSE is still used for comparison with the TimeRAF evaluation, the use of MAE and RMSE is for an interpretable scale of the PM2.5 error. The following equation (21) shows the key error measures for n observations y_j of the forecast and prediction $\hat{y}_j$.

$$\begin{aligned} \mathrm{MSE} &= \frac{1}{n}\sum_{j=1}^{n}(y_j-\hat{y}_j)^2, \\ \mathrm{MAE} &= \frac{1}{n}\sum_{j=1}^{n}|y_j-\hat{y}_j|, \\ \mathrm{RMSE} &= \sqrt{\mathrm{MSE}}, \end{aligned} \tag{21}$$

The Equation (22) is the summary of explained variance.

$$R^2 = 1-\frac{\sum_{j=1}^{n}(y_j-\hat{y}_j)^2}{\sum_{j=1}^{n}(y_j-\bar{y})^2}. \tag{22}$$

The number of forecast origins and horizons for the overall result are the same in both (21) and (22). Although included in machine-readable outputs, MAPE and sMAPE are not used for the main ranking due to the fact that percentage errors can be unstable in the case of low concentration values of PM2.5. If there are 50 or more forecast origins, then subset metrics are reported. There are 532 origins in the event subset, 293 origins in the active-event subset and 560 origins in the drift subset, and all of these pre-specified subsets meet the minimum sample-size criterion. Paired bootstrap intervals are selected from the time series as 24-hour blocks and resampled 500 times using 24-hour blocks (pairwise comparisons).

G. Comparison with Existing Studies

The direct comparison of these results with other TSFM and air-quality papers is not valid as the papers employ different sets of data, forecasting periods, variables, and evaluation schemes. For this reason, the methodological coverage is compared, in Table VII, and not cross-paper performance dominance claimed.

VIII. RESULTS AND DISCUSSION

A. Overall Comparison with Baselines

The results on the test-set are given in Table VIII. The best supervised result from the pre-specified ones is M04 which can be achieved from the PM2.5 history, weather and calendar features with MSE 26.185, MAE 3.125, RMSE 5.117, and R^2 0.379. In this run, the full feature-level Event-TimeRAF model (M09) achieves MSE 26.712 and MAE 3.149, and event-context retrieval and drift features do not improve the all-origin error over M04 in this run. A selector is not included because the result tables and prediction file of the selector do not match with the run manifest.

The frozen chronos-Bolt evaluation has been performed successfully. As a result of validation, a retrieval-fusion weight of 0.75 was chosen. On the test set, retrieval-augmented Chronos achieves an MSE of 28.709, which is approximately 0.8% better than the MSE of 28.941 given by frozen Chronos-Bolt. The corresponding difference for the paired block-bootstrap comparison is -0.233 with a 95% interval (-0.665, 0.152) and so the direction is favourable but under this uncertainty estimate it is not conclusive. MAE changes from 3.205 to 3.209, which is basically unchanged.

B. Ablation Study

Tabular comparison of the components is in Table IX. In the case of the PM2.5-only XGBoost model, the model with the clearest evidence of improvement is the one with the addition of weather and calendar information, which results in a decrease of MSE of 3.764, with the entire confidence interval lying below zero. As compared to random retrieval, cosine retrieval is an improvement in retrieval-only. For this run, however, addition of cosine retrieval features does not help in the all-origin result, which is already good on XGBoost context model. There are further no clear overall improvements in the event-hybrid retrieval and drift indicators beyond the context baseline. In this run, the interval for the comparison between M09 and the event-free A00 model also straddles the zero line and there is no confirmed overall gain from the event

TABLE VI
EXPERIMENTAL HYPERPARAMETERS AND RUNTIME SETTINGS

Component	Setting
Study period	2019–2024
Lookback and horizon	$L = 168$ hours, $H = 24$ hours
Split	70% train, 15% validation, 15% test, chronological
SACB feature groups	23 PM2.5, 14 weather, 9 calendar, 39 event features
SACB missing-data rule	Past-only filling for gaps up to 3 hours; no target interpolation
LSER retrieval	$k = 8$, sensitivity for $k \in \{1, 4, 8, 16\}$
LSER knowledge base	191 windows; 192-hour stride; training history only
LSER similarity weights	time series 0.5, weather 0.2, calendar 0.1, event 0.2
DFEH direct input	151 features for each of 24 horizon regressors
DFEH XGBoost	350 estimators, max depth 6, learning rate 0.05, subsample 0.85, colsample 0.85, $\lambda = 1.0$
DFEH drift detector	5 components; 24-hour recent window, 168-hour reference window, 0.90 validation quantile threshold
DFEH Chronos path	`amazon/chronos-bolt-small`, batch size 64
DFEH fusion weights	$\{0, 0.25, 0.5, 0.75, 1.0\}$; selected $\omega = 0.75$
Explanation record	Top 3 feature effects, windows, and recent event IDs
Bootstrap	500 paired block-bootstrap resamples, 24-hour blocks

TABLE VII
METHODOLOGICAL COMPARISON WITH RELATED FORECASTING WORK

Method family	TSFM	Retrieval	Events	Explainability	Comment
PatchTST-style Transformers [22]	No	No	No	Limited	Strong sequence modeling, but event context is not central.
Chronos [5]	Yes	No	No	Limited	Frozen zero-shot baseline used in this study.
TimeRAF [9]	Yes	Yes	No	Retrieval cases	Base retrieval-augmented TSFM method; does not target official event-aware PM2.5.
XGBoost context baseline [10]	No	No	No	Feature importance	Strong supervised tabular baseline for local PM2.5.
Event-TimeRAF	Yes	Yes	Yes	Evidence-grounded	Adds official event audit, temporal leakage control, drift indicators, and Chronos retrieval fusion.

TABLE VIII
MANIFEST-BACKED OVERALL TEST RESULTS FOR 24-HOUR PM2.5 FORECASTING

Model	MSE	MAE	RMSE	R^2
M00 persistence	47.649	4.137	6.903	-0.130
M01 daily seasonal	48.706	4.057	6.979	-0.155
M02 weekly seasonal	65.606	5.123	8.100	-0.556
M03 XGBoost PM2.5	29.949	3.368	5.473	0.290
M04 XGBoost context	**26.185**	**3.125**	**5.117**	**0.379**
M05 random retrieval	42.367	4.257	6.509	-0.005
M06 cosine retrieval	38.083	3.940	6.171	0.097
M07 XGBoost cosine	26.672	3.151	5.164	0.367
M08 Event-TimeRAF no drift	26.677	3.145	5.165	0.367
M09 Event-TimeRAF full	26.712	3.149	5.168	0.367
M10 frozen Chronos-Bolt	28.941	3.205	5.380	0.314
M11 Chronos + retrieval	28.709	3.209	5.358	0.319

TABLE IX
SELECTED PAIRED BLOCK-BOOTSTRAP ABLATION RESULTS

Comparison	MSE diff.	95% CI low	95% CI high
M04 minus M03, SACB weather/calendar	-3.764	-5.856	-2.071
M06 minus M05, LSER cosine/random ranking	-4.284	-5.862	-2.759
M07 minus M04, LSER retrieval summary	0.487	0.015	1.028
M08 minus M07, LSER event-hybrid ranking	0.005	-0.292	0.275
M09 minus M08, DFEH drift indicators	0.035	-0.101	0.161
M09 minus A00, event context	0.083	-0.225	0.354
M09 minus M04, SACB+LSER+DFEH	0.528	-0.026	1.182
M11 minus M10, DFEH Chronos fusion	-0.233	-0.665	0.152

model. The auxiliary models A00 and A01 are kept only to separate out the effects of events and random retrieval.

The ablation does not lead to a monotonous accuracy sequence: the weather-calendar context is effective, cosine

TABLE X
COSINE RETRIEVAL SENSITIVITY TO k

k	MSE	MAE	RMSE
1	65.129	5.133	8.070
4	41.237	4.134	6.422
8	38.083	3.940	6.171
16	36.488	3.851	6.040

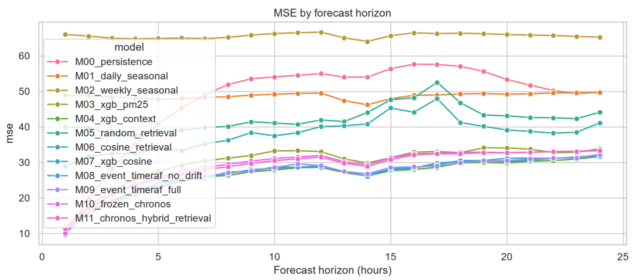

Fig. 5. MSE by forecast horizon on the Los Angeles County test set.

retrieval is effective compared to random retrieval in the retrieval-only path and the measured event and drift additions do not make the strongest of the context models. This non-monotonic behaviour is preserved, not that this means that each named piece adds accuracy.

Table X presents the results of the k sensitivity test, indicating that the retrieval only cosine baseline is better with larger number of neighbors within the range tested. The best retrieval-only configuration among k in $\{1, 4, 8, 16\}$ is k=16, with MSE 36.488. The Event-TimeRAF feature models reported below were using the default value of k=8, pre-specified for these.

C. *Event and Drift Subsets*

Representative subsets of events and drift for the strongest context model, the full Event-TimeRAF model and the two Chronos variants are given in Table XI. All subsets meet the minimum sample-size criterion which was pre-specified. These models should not be interpreted as being easier to succeed on the event subset, but rather that the event subset used for the models had lower MSE than the non-event subset for these models for the particular 2024 test-event hours kept by the audit. The drift subset is only applicable for stratified evaluation, and is defined by the DFEH flag that is validation-calibrated.

D. *Horizon-Wise and Retrieval Diagnostics*

Figure 5 and 6 depict the test errors horizon wise. Retrieval behavior is summarized in Fig. 7 and the distribution of the drift-scores over the test period is displayed in Fig. 8. The test-origin forecast for a representative test is shown as an example in figure 9, which allows M04 and M09 to be compared with the observed target. These plots were generated from the predictions identified by 20260723T112033170131Z, and contain only the M00-M11 predictions.

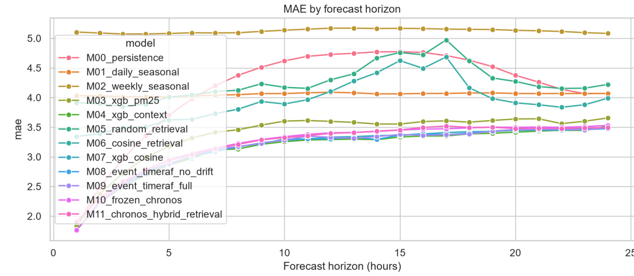

Fig. 6. MAE by forecast horizon on the Los Angeles County test set.

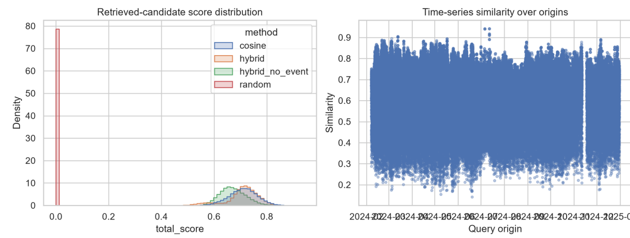

Fig. 7. Similarity and candidate diagnostics for test-set retrieval.

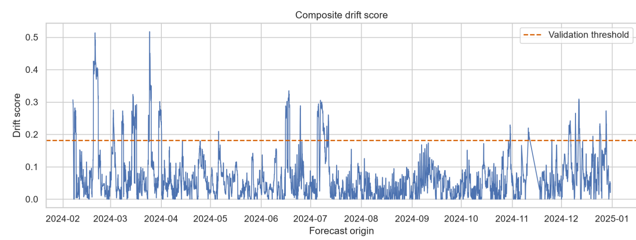

Fig. 8. Composite drift-score diagnostics on the test set.

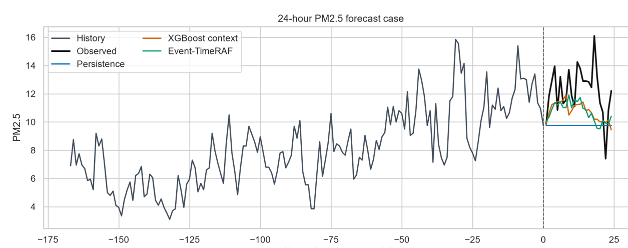

Fig. 9. Representative 24-hour forecast case from the test set.

E. *Explainability and Evidence Analysis*

Within the DFEH, explainability takes the form of a source-grounded evidence record, not as a post-hoc, free-text interpretation. The following information is stored in the pipeline for each forecast origin evaluated: Run ID, Window ID, Origin Timestamp, Retrieved Historical Evidence IDs, Event Evidence IDs, Leading Feature Effects, Drift Components, Retrieval Spread, Validation Residual MAE, Uncertainty Proxy, Drift Score, Drift Flag, and Event-availability mode. The result of the evaluation is contained in `explanations.parquet` (7,199 explanation records). A representative record contains the strongest local contribution fields (the current PM2.5 value and recent rolling PM2.5 means), links the prediction to three retrieved historical windows, links recent source-

TABLE XI
REPRESENTATIVE EVENT AND DRIFT SUBSET RESULTS

Subset	Model	Origins	MSE	MAE
Non-event	M04 XGBoost context	6,667	27.520	3.179
Event	M04 XGBoost context	532	9.451	2.450
Non-event	M09 Event-TimeRAF full	6,667	27.942	3.187
Event	M09 Event-TimeRAF full	532	11.301	2.666
Non-drift	M04 XGBoost context	6,639	26.107	3.088
Drift	M04 XGBoost context	560	27.110	3.562
Non-drift	M09 Event-TimeRAF full	6,639	26.330	3.088
Drift	M09 Event-TimeRAF full	560	31.244	3.872
Drift	M10 frozen Chronos-Bolt	560	28.333	3.476
Drift	M11 Chronos + retrieval	560	28.198	3.580

TABLE XII
EXPLANATION EVIDENCE STORED BY EVENT-TIMERAF

Evidence field	Interpretation role
Top feature effects	Local model drivers from the direct forecaster.
Retrieved windows	Historical analogues used by the retrieval module.
Event evidence IDs	Official storm-event context linked to the origin.
Drift components	Mean, variance, weather, similarity, and event-burst shift signals.
Uncertainty proxy	Combined retrieval spread and validation residual error.

TABLE XIII
COMPUTATIONAL AND REPRODUCIBILITY SUMMARY

Item	Recorded value
Run identifier	`20260723T112033170131Z`
Runtime	1,576.9 seconds
Platform	Cloud-hosted Linux environment
Python	3.12.13
Forecast horizon	24 hours
Lookback window	168 hours
Bootstrap resamples	500
Chronos checkpoint	`amazon/chronos-bolt-small`
Code availability	https://github.com/shasabbir/event-time-raf

recorded storm events when available, and marks distribution shift when the drift detector is active.

The summary of the explanation layer is given in Table XII. This design supports auditing because each explanation statement is mapped to a model feature, a retrieved time-series window, an official event record or a drift component. Event records are reported as contextual evidence rather than causal explanations, and the drift flag is reported as an uncertainty warning.

F. Computational Cost and Reproducibility

The entire experiment was carried out in a cloud-based Linux environment with Python 3.12.13 in 1,576.9 seconds. The versions of the packages recorded are: NumPy 2.0.2, pandas 2.3.3, scikit-learn 1.6.1, XGBoost 3.2.0, PyTorch 2.10.0+cu128, and chronos-forecasting 2.3.1. Artifacts checksums and configuration hash (SHA-256) is stated in the run manifest. The model of the CPU, GPU, and RAM were not recorded in the manifest and, as such, it cannot be claimed that the efficiency reported in this paper is hardware normalised in terms of the CPU, GPU, or RAM. The archived run will have the following fields (Table XIII) that were preserved for table of values reproducibility.

The overall results show that conventional supervised context features are strong for 24-hour PM2.5 forecasting in Los Angeles County. Retrieval provides an independent historical-pattern baseline and yields a small MSE improvement when combined with frozen Chronos-Bolt. However, the current feature-level Event-TimeRAF design does not outperform the best XGBoost context baseline. It should therefore be interpreted as an implemented and audited event-aware retrieval framework with selective gains rather than as a uniformly superior forecasting model. Artifacts created after the verified manifest are excluded from this interpretation.

Lastly all event aware interpretations should be read with the caveat of availability. NOAA Storm Events records are official and hash verified, and the source doesn't support getting the publication time in machine readable format. Event-related results are retrospective sensitivity results as the experiment measures the availability time as the start time for the event. The true issue or publication times of events or smoke sources would need to be used in a strict operational study, e.g. an audited NOAA HMS cache or alert product archive.

IX. LIMITATIONS

The first is the availability of the events. NOAA Storm Events records are official and are hash verified, and the public detail files don't contain machine-readable publication time. The SACB therefore uses event start as an availability proxy and the results are retrospective sensitivity analyses and not operational forecasts.

The geographic validation is the second constraint. The experiment covers one county-level PM2.5 aggregate for the period 2019-2024. It does not entail transfer to another county, monitoring topology, climate and emissions regime and thus not meet the external validation criterion for a mature generalization claim.

The third limitation is related to integration of models. The LSER is not based on a learned retriever but instead on transparent similarity functions and the DFEH is a combination of Chronos-Bolt and retrieval at forecast level. Event-TimeRAF consequently does not duplicate the learned dual encoder or built-in Channel Prompting of TimeRAF.

The fourth restriction is related to the lineage of artifacts. After archived run, a drift-gated selector was added and the edited prediction and table files no longer have the same original checksums in the manifest. The results from that selector are thus not included in the reported results. There was also no hardware model ID or peak memory reported in the run manifest and therefore the runtime is not a "hardware-normalized" efficiency value.

The fifth limitation is the cross version replay of the random-retrieval baseline. A seeded selection doesn't generate the same candidates as it does in NumPy 2.0.2 when repeated in NumPy 2.4.1. The verification notebook therefore reconstructs M05 and A01 from the original, checksum-verified retrieval-evidence rows. This maintains the reported run, but for exact random baseline retraining, one must have the software environment that was used when the run was completed.

X. Future Work

The first one to do is a clean rerun that includes any artifacts that were chosen for validation and drift gate and all artifacts downstream of the drift gate in one manifest. Then, when performing operational event studies, the event-start proxies should be replaced by archives that reveal the true time of issue or publication of the events. An external validation should use the same protocol with no changes for a second US county, having different meteorology and emissions than the first county. Future research can assess a learned event-context retriever, enhanced event embeddings, tuned probabilistic uncertainty, and built-in event-conditioning for TSFM. Those extensions should be made in a component-wise manner in the absence of any change in the source audit/temporal embargo.

XI. Conclusion

Event-TimeRAF was presented in this study for 24-hour PM2.5 forecast for Los Angeles County. It has Source-Audited Context Builder, Leakage-Safe Event-Context Retriever which enforces a strict historical embargo, and a Drift-Aware Forecast and Evidence Head which compares direct and frozen-TSFM paths and keeps traceable evidence. During the 2019-2024 test period, the best pre-specified supervised context model achieves the following results: MSE 26.185, MAE 3.125, RMSE 5.117 and R^2 0.379; the full feature-level Event-TimeRAF model achieves MSE 26.712. Retrieval shows a change in the frozen Chronos-Bolt MSE from 28.941 to 28.709, but the paired interval crosses zero. On this basis the study suggests that source-audited retrieval is a transparent method of analysis that has selective advantages, but is not necessarily the best forecaster. However, before making operational or general cross-domain claims, strict event issue times, external geographic validation and deeper retrieval integration are required.

References

[1] World Health Organization, "WHO global air quality guidelines: Particulate matter (PM2.5 and PM10), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide," 2021. [Online]. Available: https://www.who.int/publications/i/item/9789240034228

[2] A. Das, W. Kong, R. Sen, and Y. Zhou, "A decoder-only foundation model for time-series forecasting," in *Proceedings of the 41st International Conference on Machine Learning*, 2024.

[3] G. Woo, C. Liu, A. Kumar, C. Xiong, S. Savarese, and D. Sahoo, "Unified training of universal time series forecasting transformers," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235, 2024, pp. 53 140–53 164.

[4] M. Goswami, K. Szafer, A. Choudhry, Y. Cai, S. Li, and A. Dubrawski, "MOMENT: A family of open time-series foundation models," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235, 2024, pp. 16 115–16 152.

[5] A. F. Ansari, L. Stella, C. Turkmen, X. Zhang, P. Mercado, H. Shen, O. Shchur, S. S. Rangapuram, S. P. Arango, S. Kapoor, J. Zschiegner, D. C. Maddix, H. Wang, M. W. Mahoney, K. Torkkola, A. G. Wilson, M. Bohlke-Schneider, and Y. Wang, "Chronos: Learning the language of time series," *Transactions on Machine Learning Research*, 2024.

[6] V. Ekambaram, A. Jati, P. Dayama, S. Mukherjee, N. H. Nguyen, W. M. Gifford, C. Reddy, and J. Kalagnanam, "Tiny time mixers (TTMs): Fast pre-trained models for enhanced zero/few-shot forecasting of multivariate time series," in *Advances in Neural Information Processing Systems*, vol. 37, 2024.

[7] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Kuttler, M. Lewis, W. tau Yih, T. Rocktaschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9459–9474.

[8] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W. tau Yih, "Dense passage retrieval for open-domain question answering," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, 2020, pp. 6769–6781.

[9] H. Zhang, C. Xu, Y.-F. Zhang, Z. Zhang, L. Wang, and J. Bian, "TimeRAF: Retrieval-augmented foundation model for zero-shot time series forecasting," *IEEE Transactions on Knowledge and Data Engineering*, vol. 37, no. 9, pp. 5654–5665, 2025.

[10] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.

[11] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[12] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[13] K. Cho, B. van Merrienboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, "Learning phrase representations using RNN encoder–decoder for statistical machine translation," in *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*, 2014, pp. 1724–1734.

[14] R. Godahewa, C. Bergmeir, G. I. Webb, R. J. Hyndman, and P. Montero-Manso, "Monash time series forecasting archive," in *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021.

[15] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang, "Informer: Beyond efficient transformer for long sequence time-series forecasting," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, no. 12, 2021, pp. 11 106–11 115.

[16] H. Wu, J. Xu, J. Wang, and M. Long, "Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting," in *Advances in Neural Information Processing Systems*, vol. 34, 2021.

[17] T. Zhou, Z. Ma, Q. Wen, X. Wang, L. Sun, and R. Jin, "FEDformer: Frequency enhanced decomposed transformer for long-term series forecasting," in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162, 2022, pp. 27 268–27 286.

[18] Y. Zhang and J. Yan, "Crossformer: Transformer utilizing cross-dimension dependency for multivariate time series forecasting," in *Proceedings of the 11th International Conference on Learning Representations*, 2023.

[19] Y. Liu, T. Hu, H. Zhang, H. Wu, S. Wang, L. Ma, and M. Long, "iTransformer: Inverted transformers are effective for time series forecasting," in *Proceedings of the 12th International Conference on Learning Representations*, 2024.

[20] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, "TimesNet: Temporal 2d-variation modeling for general time series analysis," in *Proceedings of the 11th International Conference on Learning Representations*, 2023.

[21] A. Zeng, M. Chen, L. Zhang, and Q. Xu, "Are transformers effective for time series forecasting?" in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 9, 2023, pp. 11 121–11 128.

[22] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam, "A time series is worth 64 words: Long-term forecasting with transformers," in *Proceedings of the 11th International Conference on Learning Representations*, 2023.

[23] S. Wang, H. Wu, X. Shi, T. Hu, H. Luo, L. Ma, J. Y. Zhang, and J. Zhou, "TimeMixer: Decomposable multiscale mixing for time series forecasting," in *Proceedings of the 12th International Conference on Learning Representations*, 2024.

[24] Y. Liu, H. Zhang, C. Li, X. Huang, J. Wang, and M. Long, "Timer: Generative pre-trained transformers are large time series models," in *Proceedings of the 41st International Conference on Machine Learning*, 2024, pp. 32 369–32 399.

[25] Y. Liu, G. Qin, X. Huang, J. Wang, and M. Long, "Timer-XL: Long-context transformers for unified time series forecasting," in *Proceedings of the 13th International Conference on Learning Representations*, 2025.

[26] X. Liu, J. Hu, Y. Li, S. Diao, Y. Liang, B. Hooi, and R. Zimmermann, "UniTime: A language-empowered unified model for cross-domain time series forecasting," in *Proceedings of the ACM Web Conference*, 2024, pp. 4095–4106.

[27] M. Jin, S. Wang, L. Ma, Z. Chu, J. Y. Zhang, X. Shi, P.-Y. Chen, Y. Liang, Y.-F. Li, S. Pan, and Q. Wen, "Time-LLM: Time series forecasting by reprogramming large language models," in *Proceedings of the 12th International Conference on Learning Representations*, 2024.

[28] T. Zhou, P. Niu, X. Wang, L. Sun, and R. Jin, "One fits all: Power general time series analysis by pretrained LM," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.

[29] J. Liu, L. Yang, H. Li, and S. Hong, "Retrieval-augmented diffusion models for time series forecasting," in *Advances in Neural Information Processing Systems*, 2024, pp. 2766–2786.

[30] Y. Ang, Y. Bao, Q. Huang, A. K. H. Tung, and Z. Huang, "TSGAssist: An interactive assistant harnessing LLMs and RAG for time series generation recommendations and benchmarking," *Proceedings of the VLDB Endowment*, vol. 17, no. 12, pp. 4309–4312, 2024.

[31] S. Zhou, W. Wang, L. Zhu, Q. Qiao, and Y. Kang, "Deep-learning architecture for PM2.5 concentration prediction: A review," *Environmental Science and Ecotechnology*, vol. 21, p. 100400, 2024.

[32] Z. Zhang and S. Zhang, "Modeling air quality PM2.5 forecasting using deep sparse attention-based transformer networks," *International Journal of Environmental Science and Technology*, vol. 20, pp. 13 535–13 550, 2023.

[33] Y. Liang, Y. Xia, S. Ke, Y. Wang, Q. Wen, J. Zhang, Y. Zheng, and R. Zimmermann, "AirFormer: Predicting nationwide air quality in china with transformers," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 12, 2023, pp. 14 329–14 337.

[34] US Environmental Protection Agency, "AirData: Pre-generated air quality data files," https://aqs.epa.gov/aqsweb/airdata/download_files.html, 2026, accessed: 2026-07-17.

[35] NOAA National Centers for Environmental Information, "Global hourly—integrated surface database," https://www.ncei.noaa.gov/products/land-based-station/integrated-surface-database, 2026, accessed: 2026-07-17.

[36] ——, "Storm events database: Bulk data download," https://www.ncei.noaa.gov/stormevents/ftp.jsp, 2026, accessed: 2026-07-17.

[37] NOAA Office of Satellite and Product Operations, "Hazard mapping system fire and smoke product," https://ospo.noaa.gov/products/land/hms.html, 2026, accessed: 2026-07-17.

[38] Z. Liu, M. Cheng, Z. Li, Z. Huang, Q. Liu, Y. Xie, and E. Chen, "Adaptive normalization for non-stationary time series forecasting: A temporal slice perspective," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.

[39] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[40] C. Raman, A. Nonnemaker, A. Villegas-Morcillo, H. Hung, and M. Loog, "Why did this model forecast this future? information-theoretic saliency for counterfactual explanations of probabilistic regression models," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.